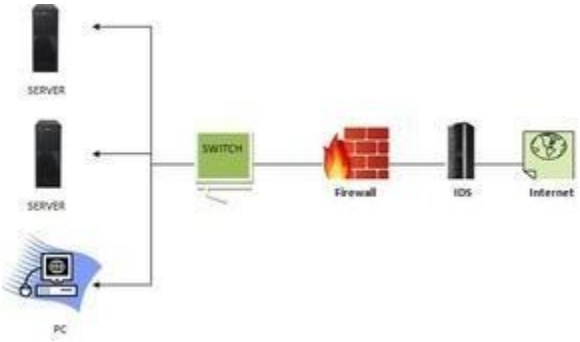
http

UNIT IV INTRUSION DETECTION

Host -Based Intrusion Detection – Network -Based Intrusion Detection – Distributed or Hybrid Intrusion Detection – Intrusion Detection Exchange Format – Honeypots – Example System Snort.

Intrusion Detection System (IDS)

A system called an intrusion detection system (IDS) observes network traffic for malicious transactions and sends immediate alerts when it is observed. It is software that checks a network or system for malicious activities or policy violations. Each illegal activity or violation is often recorded either centrally using a SIEM system or notified to an administration. IDS monitors a network or system for malicious activity and protects a computer network from unauthorized access from users, including perhaps insiders. The intrusion detector learning task is to build a predictive model (i.e. a classifier) capable of distinguishing between 'bad connections' (intrusion/attacks) and 'good (normal) connections'.



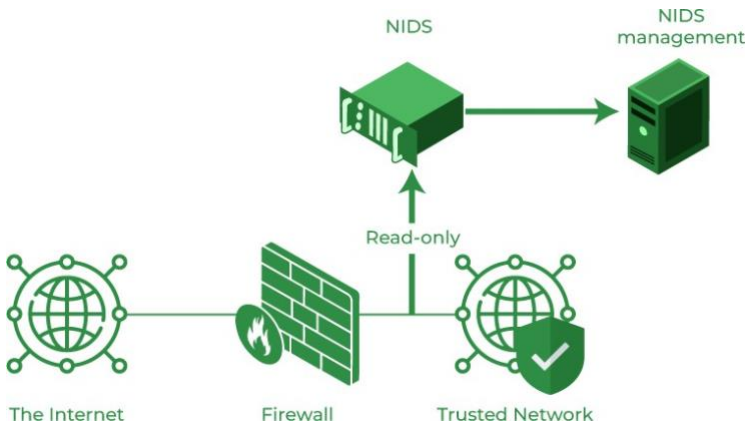
How does an IDS work?

- An IDS (Intrusion Detection System) monitors the traffic on a computer network to detect any suspicious activity.
- It analyzes the data flowing through the network to look for patterns and signs of abnormal behavior.
- The IDS compares the network activity to a set of predefined rules and patterns to identify any activity that might indicate an attack or intrusion.
- If the IDS detects something that matches one of these rules or patterns, it sends an alert to the system administrator.
- The system administrator can then investigate the alert and take action to prevent any damage or further intrusion.

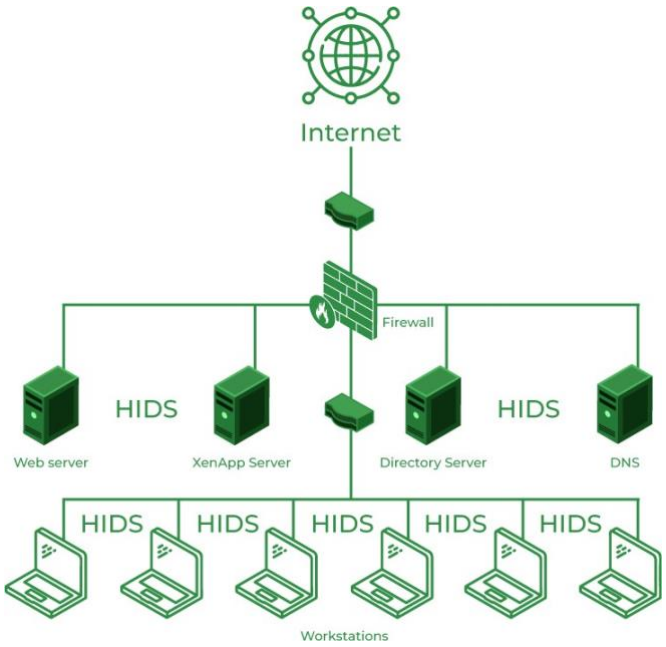
Classification of Intrusion Detection System

IDS are classified into 5 types:

- **Network Intrusion Detection System (NIDS):** Network intrusion detection systems (NIDS) are set up at a planned point within the network to examine traffic from all devices on the network. It performs an observation of passing traffic on the entire subnet and matches the traffic that is passed on the subnets to the collection of known attacks. Once an attack is identified or abnormal behavior is observed, the alert can be sent to the administrator. An example of a NIDS is installing it on the subnet where firewalls are located in order to see if someone is trying to crack the firewall.



- **Host Intrusion Detection System (HIDS):** Host intrusion detection systems (HIDS) run on independent hosts or devices on the network. A HIDS monitors the incoming and outgoing packets from the device only and will alert the administrator if suspicious or malicious activity is detected. It takes a snapshot of existing system files and compares it with the previous snapshot. If the analytical system files were edited or deleted, an alert is sent to the administrator to investigate. An example of HIDS usage can be seen on mission-critical machines, which are not expected to change their layout.



- **Protocol-based Intrusion Detection System (PIDS):** Protocol-based intrusion detection system (PIDS) comprises a system or agent that would consistently reside at the front end of a server, controlling and interpreting the protocol between a user/device and the server. It is trying to secure the web server by regularly monitoring the HTTPS protocol stream and accepting the related HTTP protocol. As HTTPS is unencrypted and before instantly entering its web presentation layer then this system would need to reside in this interface, between to use the HTTPS.

- **Application Protocol-based Intrusion Detection System (APIDS):** An application Protocol-based Intrusion Detection System (APIDS) is a system or agent that generally resides within a group of servers. It identifies the intrusions by monitoring and interpreting the communication on application-specific protocols. For example, this would monitor the SQL protocol explicitly to the middleware as it transacts with the database in the web server.
- **Hybrid Intrusion Detection System:** Hybrid intrusion detection system is made by the combination of two or more approaches to the intrusion detection system. In the hybrid intrusion detection system, the host agent or system data is combined with network information to develop a complete view of the network system. The hybrid intrusion detection system is more effective in comparison to the other intrusion detection system. Prelude is an example of Hybrid IDS.

Detection Method of IDS

1. **Signature-based Method:** Signature-based IDS detects the attacks on the basis of the specific patterns such as the number of bytes or a number of 1s or the number of 0s in the network traffic. It also detects on the basis of the already known malicious instruction sequence that is used by the malware. The detected patterns in the IDS are known as signatures. Signature-based IDS can easily detect the attacks whose pattern (signature) already exists in the system but it is quite difficult to detect new malware attacks as their pattern (signature) is not known.
2. **Anomaly-based Method:** Anomaly-based IDS was introduced to detect unknown malware attacks as new malware is developed rapidly. In anomaly-based IDS there is the use of machine learning to create a trustful activity model and anything coming is compared with that model and it is declared suspicious if it is not found in the model. The machine learning- based method has a better-generalized property in comparison to signature-based IDS as these models can be trained according to the applications and hardware configurations.

4.1 **HIDS (Host-Based Intrusion Detection System)**

A Host-Based Intrusion Detection System, or HIDS, is a type of cybersecurity solution that monitors IT systems for signs of suspicious activity to detect unusual behaviors or patterns associated either with human users or applications that could be a sign of a security breach or attempted attack.

HIDS systems are so-named because they operate on individual host systems. In this context, a host could be a server, a PC, or any other type of device that produces logs, metrics, and other data that can be monitored for security purposes.

Keep reading for everything you need to know about what a HIDS is, how HIDS solutions work, and how HIDS compares to other types of security tools.

HIDS vs. NIDS

Host-Based Intrusion Detection Systems are similar in some ways to Network Intrusion Detection Systems, or NIDS, but they are not the same type of solution.

A NIDS monitors for suspicious activity from the perspective of the network, using data sources like network switch logs. By analyzing this data, a NIDS can look for suspicious activity.

A HIDS may also monitor network activity, but it does so from the perspective of individual hosts, not centralized networking equipment like switches. In addition, for a HIDS, network data is just one of many data sources used for security analysis purposes.

How does Host-Based Intrusion Detection work?

Host-Based Intrusion Detection works by collecting data from servers, computers, and other host systems, then analyzing the data for anomalies or suspicious activity.

The data that HIDS tools analyze may include security-centric data sources, such as authentication logs (which record login events). However, a HIDS typically also analyzes other types of data, like application and operating system logs. Even though the latter types of data are not related to security specifically, unusual patterns within those data sets could be linked to security issues.

For example, a HIDS could monitor network traffic flows to detect that an application has suddenly begun receiving high volumes of requests from previously unknown external IP addresses. This activity could be the sign of a brute-force login attempt or an effort to probe the application for vulnerabilities that attackers could exploit. With this information, security teams could block the offending IP addresses.

To deliver results that are as accurate as possible, a HIDS should link and correlate different types of data sources, which makes it possible to gain deeper context on potential security events.

For instance, network traffic logs can be analyzed in conjunction with application event logs so that the HIDS can determine whether unusual activity on the network correlates with unusual activity by the application.

In the former case, it's possible that attackers are trying to find a vulnerability in the application but have not yet succeeded. In the latter, it's possible that they have breached the application, which is why the HIDS detects anomalous behavior by the application as well as unusual traffic patterns on the network.

Types of HIDS

Host-Based Intrusion Detection Systems can be broken into two main categories based on how they are deployed:

- **Agent-based HIDS**: An agent-based HIDS relies on software agents that are installed on each host to collect information from the host. This is a "heavier-weight" approach because running agents on hosts increases the resource utilization of the hosts.
- **Agentless HIDS**: With an agentless HIDS, information from hosts is collected without relying on agents, such as by streaming the data over the network. This type of HIDS is more complex to implement, and agentless HIDS sometimes can't access as much data as agent- based solutions, but the agentless approach offers the benefit of consuming fewer resources.

HIDS components

No matter which type of HIDS you deploy, your HIDS solution will typically include three main components:

- **Data collectors**: Using either agents or an agentless approach, your HIDS deploys sensors that collect data from hosts.
- **Data storage**: After being collected, the data is usually aggregated and stored in a central location. The data is retained at least as long as is necessary to analyze it, although organizations may also choose to keep the data on hand so they can reference it at a later time if desired.

- **Analytics engine**: The HIDS uses an analytics engine to process and evaluate the various data sources that it collects. The purpose of analytics is to look for patterns or anomalies, then assess the likelihood that they are the result of security risks or attacks.

HIDS capabilities

After a HIDS detects potential security problems, it can do three main things.

Alerting

The first is alerting. Alerting is the process of informing IT and/or security teams about a potential security issue.

Ideally, HIDS alerting features should be capable of assessing the severity of each security risk the HIDS identifies, then generating alerts accordingly. For example, low-risk security events should be labeled as such so that engineers are aware that those alerts are not likely to require immediate attention.

Reporting

HIDS platforms can generate reports about the overall state of security within an IT environment. The data included in reports can vary, but it may include the number and types of security risks identified by a HIDS over time, for instance, or how security issues vary across different types of hosts (such as Windows-based hosts versus Linux-based systems).

Reporting is useful for assessing security trends over time, as well as for demonstrating the security posture of an organization.

Response

In some cases, HIDS tools are capable of carrying out certain automated response activities to help remediate risks. For example, if a HIDS determines that a particular external endpoint is trying to probe a company's servers, it could automatically generate firewall rules to block the probes.
Automated remediation like this not only saves time and effort on the part of engineers, but also ensures that security risks can be blocked immediately.

HIDS security considerations and best practices

To get the most value out of a HIDS, consider best practices like the following:

- **Monitor all hosts**: A HIDS is of limited value if it only monitors some hosts. To gain the broadest possible context on security risks, your HIDS should monitor all hosts. That way, you'll know whether and how quickly security issues spread among hosts, as well as how many of your hosts are targeted by attacks. You'll also be able to detect attacks that target just one host rather than attempts to reach many hosts at once.

- **Contextualize data**: As noted above, the more data sources your HIDS analyzes collectively, the greater the context it has on potential security risks. Context is critical for distinguishing actual risks from false positives and generating accurate alerts.

- **Configure smart alerts**: To avoid distracting engineers with "alert fatigue," a HIDS should be configured to alert only on events that require a response. Alerts should also be categorized based on severity level so that engineers know which ones to prioritize.

- **Consider agentless HIDS**: While an agent-based HIDS has its advantages (such as easier access to host-based data), agentless HIDS solutions are easier to deploy and manage in many

respects because they don't require installing software agents on each host. They are also lighter on resource consumption.

These practices help ensure that your HIDS delivers the most actionable security insights at the lowest total cost.

Limitations and challenges of HIDS

While a HIDS provides one layer of defense against security threats, it's just that – one layer.

Since HIDS platforms specialize only in host-based security, they're of limited use for addressing other types of threats, such as vulnerabilities within application source code or public cloud workloads.

That's why it's important to deploy additional security tools and services alongside a HIDS in order to gain comprehensive protection against potential security risks across all facets of your environment.

4.2 NIDS(Network Intrusion Detection System)

NIDS is a security tool designed to detect, monitor, and analyze traffic for suspicious activity or malicious attacks. It is essential to a larger security infrastructure and prevents network breaches and data theft.

As mentioned, NIDS (Network Intrusion Detection System) is a security technology that monitors and analyzes network traffic for signs of malicious activity, unauthorized access, or security policy violations. The primary function of a NIDS is to detect and alert network administrators of any potential or ongoing attacks on the network.

NIDS works by examining data packets for specific patterns and behaviors that indicate the presence of an attack. It can detect and alert network administrators of attacks such as DoS (Denial of Service), port scanning, virus and malware infections, and unauthorized access attempts.

NIDS is an essential component of a comprehensive network security strategy. It helps to identify and respond to threats quickly before they can cause significant damage or compromise sensitive data.

How Does NIDS Work?

Network-based Intrusion Detection System analyzes the network traffic and looks for behavior patterns indicative of an intrusion or attack. It typically operates in a passive or inline mode, and they use different detection methods to identify network intrusions.

In passive mode, the NIDS monitors outgoing network traffic without interfering with it. In inline mode, the NIDS can modify network traffic to detect intrusions or block malicious activities.
However, the active mode may increase the risk of disrupting legitimate network traffic, and it is usually not recommended.

When a NIDS detects a potential network threat, it generates an alert. The alert includes information such as the type of attack, the source and destination IP addresses, and the time of the attack. The NIDS may also take action to prevent the attack, such as blocking the source IP address or modifying current network traffic.

Methods of NIDS Detection

Network Intrusion Detection Systems are designed to detect network-based attacks and intrusions. They use different detection methods to identify suspicious traffic and abnormal behavior. There are

three primary detection methods used by NIDS: signature-based detection, anomaly-based detection, and hybrid detection.

1. Signature-Based Detection

This method compares traffic passing through the network against known attack signatures or patterns. Attack signatures are predefined network traffic patterns associated with specific types of attacks.

The NIDS alerts the network administrator if the traffic matches a known signature. Signature-based detection is effective at identifying known attacks, but it cannot detect new or unknown attacks.

2. Anomaly-Based Detection

This method involves detecting traffic that deviates from the normal network behavior. NIDS monitors network traffic and generates an alert if it detects any activity outside the expected range. Anomaly-based detection is useful in detecting new or unknown attacks but can generate many false positives.

3. Hybrid Detection

This method combines signature-based and anomaly-based detection methods. The NIDS first uses signature-based detection to identify known attacks and then anomaly-based detection to identify unknown attacks. By combining both methods, hybrid detection can provide high accuracy and minimize the false positive rate.

Others

In addition to these three primary methods, NIDS can use other techniques, such as protocol and heuristic analysis. Protocol analysis involves examining network traffic to detect protocol violations and abnormal behavior. The heuristic analysis involves identifying patterns of behavior that are associated with attacks.

Technologies That a Network-Based Intrusion Detection System Can Monitor

NIDS systems can monitor network technologies and protocols to detect potential security breaches. Here are some of the technologies that these systems can monitor:

1. Network Protocols

NIDS systems can monitor network protocols such as TCP/IP, HTTP, FTP, DNS, SMTP, and SNMP to detect anomalous behavior that might indicate a network attack. For example, the system can detect any attempts to exploit vulnerabilities in the protocol to gain unauthorized access.

2. Network Devices

NIDS systems can monitor network devices such as routers, switches, and firewalls to detect unauthorized access or configuration changes. The system can also detect any attempts to exploit vulnerabilities in the devices to gain access to the network.

3. Applications

NIDS systems can monitor network applications such as email servers, web servers, and databases to detect any unusual activity that might indicate a security breach. For example, the system can detect attempts to access sensitive information or execute malicious code.

4. Operating Systems

NIDS systems can monitor the operating systems of network devices and servers to detect any security vulnerabilities or malicious activity. The system can detect any attempts to exploit vulnerabilities in the operating system to gain unauthorized access.

5. Wireless Networks

NIDS can monitor wireless networks to detect any unauthorized access or malicious activities. The system can monitor the wireless traffic and identify rogue access points, unauthorized connections, or denial of service attacks.

Advantages of Network Intrusion Detection System

Network Intrusion Detection Systems (NIDS) are essential to network security infrastructure. Here are some of the vital advantages of using NIDS:

1. Prevention of Network Attacks

NIDS actively monitors the network traffic for any suspicious activities and potential threats. It can detect and block any unauthorized attempts to access the network, such as port scanning, password guessing, and other attacks. By preventing these attacks, NIDS can help maintain network security and prevent data breaches.

2. Identification of Vulnerabilities

NIDS can scan for vulnerabilities in the network, such as misconfigured devices, outdated software, and unsecured network connections. Once these vulnerabilities are detected, they can be addressed before attackers can exploit them, preventing potential security breaches.

3. Protection of Sensitive Information

NIDS can help protect sensitive information, such as customer data, financial records, and intellectual property, by monitoring the network for any unauthorized access attempts. If an attempt is detected, NIDS can alert security personnel, who can take appropriate action to prevent data loss or theft.

4. Real-Time Monitoring

NIDS provides real-time network monitoring, allowing security personnel to respond to any threats or attacks quickly. This quick response can help prevent any potential damage caused by the attack and minimize downtime.

5. Compliance with Regulations

NIDS can help organizations comply with various regulations such as HIPAA, PCI-DSS, and GDPR, which require organizations to have proper security measures to protect sensitive data.

Limitations of Network Intrusion Detection Systems

1. Need for Frequent Updating

This is because new attack methods are constantly being developed, and NIDS must be able to detect these new threats. NIDS typically uses signature-based detection methods, which must be updated with new signatures to detect new attacks. If the system is not updated regularly, it may miss new threats.

2. Time-Consuming Process

NIDS requires extensive configuration to ensure it is tailored to the organization’s needs. The configuration of NIDS includes defining the types of traffic that should be monitored, setting the detection thresholds, and configuring the alerting and reporting mechanisms.

This can be time-consuming and requires a skilled technician to ensure the system is optimized for the organization's needs.

3. Regular Maintenance

NIDS requires maintenance to ensure that it is functioning properly. This includes monitoring the system to ensure it generates alerts correctly, responds promptly, and addresses any issues. Regular maintenance is essential to ensure that the system functions at peak performance and provides the level of protection the organization requires.

Who is NIDS For?

Network Intrusion Detection Systems are for any individual, organization, or business that needs to ensure the security of their network. This includes government agencies, large and small businesses, educational institutions, and individuals with sensitive information on their networks.

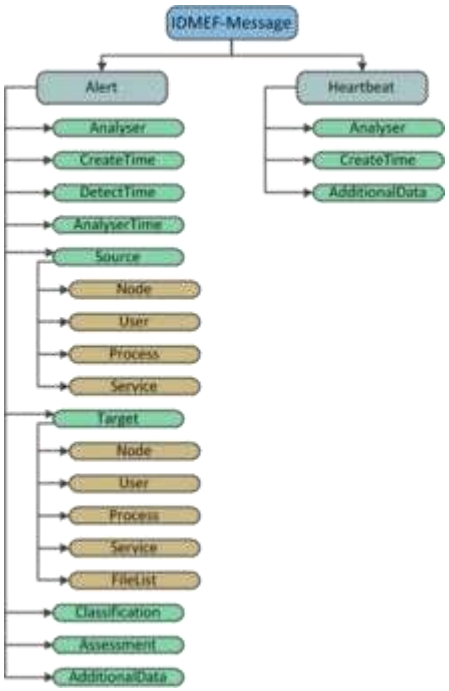
NIDS is essential for organizations that handle sensitive data, such as personal, financial, or confidential business information. It can detect and alert system administrators to potential threats, allowing them to take action to prevent or minimize damage to the network.

Moreover, NIDS can be used by security professionals, network administrators, and IT teams to monitor network traffic and identify potential security issues before they can cause harm. It can also help with compliance requirements for certain industries, such as healthcare or financial services, which are required to protect sensitive data.

Intrusion Detection Message Exchange Format

Used as part of computer security, **IDMEF** (**Intrusion Detection Message Exchange Format**) is a data format used to exchange information between software enabling intrusion detection, intrusion prevention, security information collection and management systems that may need to interact with them. IDMEF messages are designed to be processed automatically. The details of the format are described in the RFC 4765. This RFC presents an implementation of the XML data model and the associated DTD. The requirements for this format are described in RFC 4766, and the recommended transport protocol (IDXP) is documented in RFC 4767

IDMEF



The purpose of IDMEF is to define data formats and exchange procedures for sharing information of interest to intrusion detection and response systems and to the management systems that may need to interact with them. It is used in computer security for incidents reporting and exchanging. It is intended for easy automatic processing.

IDMEF is a well-structured object-oriented format, which consists of 33 classes containing 108 fields, including three mandatory:

- The classification
- The unique login
- The date of creation of the alert.

There are currently two types of IDMEF messages that can be created, Heartbeat or Alert

Heartbeat

The Heartbeats are sent by the analyzers to indicate their status. These messages are sent at regular intervals which period is defined in the Heartbeat Interval Field. If none of these messages are received for several periods of time, consider that this analyzer is not able to trigger alerts.

Alert

Alerts are used to describe an attack that took place, the main areas that create the alert are:

- **CreateTime**: Date of creation of the alert
- **DetectTime**: alert detection time by the analyzer
- **AnalyzerTime**: The time the alert was sent by the analyzer
- **Source**: Details about the origin of the attack can be a service, a user, a process and / or a node
- **Target**: Details on the target of the attack can be a service, a user, a process and / or a node and a file

- **Classification**: Name of the attack and references, as CVEs
- **Assessment**: Evaluation of the attack (severity, potential impact, etc.)
- **AdditionalData**: Additional information on the attack

There are three other alert types that inherit from this scheme:

- **CorrelationAlert**: Grouping of alerts related to one another
- **ToolAlert**: alerts from the same Grouping tool
- **OverflowAlert**: Alert resulting from attack so-called buffer overflow

4.3 Distributed or Hybrid Intrusion Detection System (DIDS)

A **Distributed Intrusion Detection System (DIDS)** or **Hybrid Intrusion Detection System** combines aspects of **Host-Based Intrusion Detection Systems (HIDS)** and **Network-Based Intrusion Detection Systems (NIDS)** into a cohesive, interconnected solution. It leverages distributed agents, sensors, or nodes to monitor multiple points in a network or infrastructure, offering broader and more detailed coverage against potential threats.

Key Characteristics

1. **Integration of Multiple IDS Types**:
   - Combines HIDS and NIDS to provide a comprehensive view of system and network activities.
2. **Distributed Monitoring**:
   - Deploys sensors or agents across various hosts, network devices, or locations.
3. **Centralized Analysis**:
   - Aggregates data from distributed agents to a central analysis server or control node.
   - Correlates events across different components for holistic threat detection.
4. **Real-Time Detection**:
   - Provides real-time or near-real-time alerts by analyzing data as it is collected.
5. **Scalability**:
   - Suitable for large, distributed environments such as enterprise networks, cloud infrastructures, and IoT ecosystems.

Components of DIDS

1. **Sensors or Agents**:
   - Deployed on hosts (HIDS) or network devices (NIDS) to collect data.
   - Monitor logs, traffic, or system activities locally.
2. **Communication Layer**:
   - Ensures secure and efficient data transfer between distributed sensors and the central server.
3. **Centralized Server**:
   - Aggregates and correlates data from distributed sensors.
   - Hosts analysis tools and machine learning models for detecting threats.
4. **Management Console**:
   - Provides administrators with an interface to configure, monitor, and analyze IDS data.

Advantages of DIDS

1. **Comprehensive Coverage**:
   - Monitors both network-wide and host-specific activities, reducing blind spots.
2. **Enhanced Detection Capabilities**:
   - Correlates data from multiple sources to identify sophisticated or multi-vector attacks.
3. **Scalability**:
   - Easily adaptable to growing or geographically dispersed networks.
4. **Resilience**:
   - If one sensor is compromised, others continue functioning, maintaining monitoring capabilities.
5. **Customization**:
   - Supports tailored configurations for specific network segments or hosts.

Challenges of DIDS

1. **Complex Deployment**:
   - Requires careful planning and configuration for distributed environments.
2. **Resource-Intensive**:
   - Aggregating and analyzing large volumes of data can strain resources.
3. **Latency in Detection**:
   - Centralized analysis might introduce slight delays in detecting threats.
4. **Potential Single Point of Failure**:
   - If the central server is compromised or fails, it could affect the entire system.
5. **High Cost**:
   - May require significant investment in infrastructure, software, and skilled personnel.

Use Cases for DIDS

1. **Large Enterprise Networks**:
   - Provides coverage across multiple departments, data centers, and remote offices.
2. **Cloud Environments**:
   - Monitors activities across hybrid or multi-cloud infrastructures.
3. **Critical Infrastructure**:
   - Ensures security for SCADA systems, smart grids, and industrial control systems.
4. **IoT Networks**:
   - Secures a vast array of interconnected devices with diverse communication protocols.

Popular DIDS Solutions

1. **Snort + OSSEC**:
   - A combination of Snort (NIDS) and OSSEC (HIDS) for hybrid detection.
2. **Wazuh**:
   - Provides integrated HIDS, NIDS, and SIEM (Security Information and Event Management) capabilities.
3. **Suricata + Zeek (formerly Bro)**:
   - Network intrusion detection combined with traffic analysis and forensic capabilities.
4. **IBM QRadar**:
   - A unified security platform with DIDS capabilities.

4.4 Honeypot

Honeypot is a network-attached system used as **a trap for cyber-attackers** to detect and study the tricks and types of attacks used by hackers. It acts as a potential target on the internet and informs the defenders about any unauthorized attempt to the information system.

Honeypots are mostly used by large companies and organizations involved in cybersecurity. It helps cybersecurity researchers to learn about the different type of attacks used by attackers. It is suspected that even the cybercriminals use these honeypots to decoy researchers and spread wrong information.

The **cost of a honeypot** is generally **high** because it requires specialized skills and resources to implement a system such that it appears to provide an organization's resources still preventing attacks at the backend and access to any production system.

A **honeynet** is a combination of two or more honeypots on a network.

Types of Honeypot:

Honeypots are classified based on their deployment and the involvement of the intruder.
Based on their deployment, honeypots are divided into :

1. **Research honeypots-** These are used by researchers to analyze hacker attacks and deploy different ways to prevent these attacks.
2. **Production honeypots-** Production honeypots are deployed in production networks along with the server. These honeypots act as a frontend trap for the attackers, consisting of false information and giving time to the administrators to improve any vulnerability in the actual system.

Based on interaction, honeypots are classified into:

1. **Low interaction honeypots:**Low interaction honeypots gives very little insight and control to the hacker about the network. It simulates only the services that are frequently requested by the attackers. The main operating system is not involved in the low interaction systems and therefore it is less risky. They require very fewer resources and are easy to deploy. The only disadvantage of these honeypots lies in the fact that experienced hackers can easily identify these honeypots and can avoid it.
2. **Medium Interaction Honeypots:** Medium interaction honeypots allows more activities to the hacker as compared to the low interaction honeypots. They can expect certain activities and are designed to give certain responses beyond what a low-interaction honeypot would give.

3. **High Interaction honeypots:**A high interaction honeypot offers a large no. of services and activities to the hacker, therefore, wasting the time of the hackers and trying to get complete information about the hackers. These honeypots involve the real-time operating system and therefore are comparatively risky if a hacker identifies the honeypot. High interaction honeypots are also very costly and are complex to implement. But it provides us with extensively large information about hackers.

Advantages of honeypot:

1. Acts as a rich source of information and helps collect real-time data.
2. Identifies malicious activity even if encryption is used.
3. Wastes hackers' time and resources.
4. Improves security.

Disadvantages of honeypot:

1. Being distinguishable from production systems, it can be easily identified by experienced attackers.
2. Having a narrow field of view, it can only identify direct attacks.
3. A honeypot once attacked can be used to attack other systems.
4. Fingerprinting(an attacker can identify the true identity of a honeypot).

Types of Honeypots

Honeypots can also be broken down by the type of activity they detect.

Email trap or spam trap

An email or spam trap will implant a fictitious email address in a hidden field that can only be detected by an automated address harvester or site crawler. Since the address is not visible to legitimate users, the organization can categorize all correspondence delivered to that inbox as spam. The organization can then block that sender and its IP address, as well as any messages that match its content.

Decoy Database

A decoy database is an intentionally vulnerable fictitious data set that helps organizations monitor software vulnerabilities, architecture insecurities or even nefarious internal actors. The decoy database will gather information about injection techniques, credential hijacking or privilege abuse used by an attacker that can then be built into system defenses and security policies.

Malware Honeypot

A malware honeypot mimics a software app or an application programming interface (API) in an attempt to draw out malware attacks in a controlled, non-threatening environment. In doing so, the infosec team can then analyze the attack techniques and develop or enhance anti-malware solutions to address these specific vulnerabilities, threats or actors.

Spider Honeypot

Similar to the spam honeypot, a spider honeypot is designed to trap web crawlers, sometimes called spiders, by creating web pages and links only accessible to automated crawlers. Identifying these spiders can help organizations understand how to block malicious bots, as well as ad-network crawlers.

4.5 SNORT Definition

SNORT is a powerful open-source intrusion detection system (IDS) and intrusion prevention system (IPS) that provides real-time network traffic analysis and data packet logging. SNORT uses a rule-based language that combines anomaly, protocol, and signature inspection methods to detect potentially malicious activity.

Using SNORT, network admins can spot denial-of-service (DoS) attacks and distributed DoS (DDoS) attacks, Common Gateway Interface (CGI) attacks, buffer overflows, and stealth port scans. SNORT creates a series of rules that define malicious network activity, identify malicious packets, and send alerts to users.

SNORT is a free-to-use open-source piece of software that can be deployed by individuals and organizations. The SNORT rule language determines which network traffic should be collected and what should happen when it detects malicious packets. This snorting meaning can be used in the same way as sniffers and network intrusion detection systems to discover malicious packets or as a full network IPS solution that monitors network activity and detects and blocks potential attack vectors.

What Are the Features of SNORT?

There are various features that make SNORT useful for network admins to monitor their systems and detect malicious activity. These include:

Real-time Traffic Monitor

SNORT can be used to monitor the traffic that goes in and out of a network. It will monitor traffic in real time and issue alerts to users when it discovers potentially malicious packets or threats on Internet Protocol (IP) networks.

Packet Logging

SNORT enables packet logging through its packet logger mode, which means it logs packets to the disk. In this mode, SNORT collects every packet and logs it in a hierarchical directory based on the host network's IP address.

Analysis of Protocol

SNORT can perform protocol analysis, which is a network sniffing process that captures data in protocol layers for additional analysis. This enables the network admin to further examine potentially malicious data packets, which is crucial in, for example, Transmission Control Protocol/IP (TCP/IP) stack protocol specification.

Content Matching

SNORT collates rules by the protocol, such as IP and TCP, then by ports, and then by those with content and those without. Rules that do have content use a multi-pattern matcher that increases performance, especially when it comes to protocols like the Hypertext Transfer Protocol (HTTP). Rules that do not have content are always evaluated, which negatively affects performance.

OS Fingerprinting

Operating system (OS) fingerprinting uses the concept that all platforms have a unique TCP/IP stack. Through this process, SNORT can be used to determine the OS platform being used by a system that accesses a network.

Can Be Installed in Any Network Environment

SNORT can be deployed on all operating systems, including Linux and Windows, and as part of all network environments.

Open Source

As a piece of open-source software, SNORT is free and available for anyone who wants to use an IDS or IPS to monitor and protect their network.

Rules Are Easy to Implement

SNORT rules are easy to implement and get network monitoring and protection up and running. Its rule language is also very flexible, and creating new rules is pretty simple, enabling network admins to differentiate regular internet activity from anomalous or malicious activity.

What Are the Different SNORT Modes?

There are three different modes that SNORT can be run in, which will be dependent on the flags used in the SNORT command.

Packet Sniffer

SNORT's packet sniffer mode means the software will read IP packets then display them to the user on its console.

Packet Logger

In packet logger mode, SNORT will log all IP packets that visit the network. The network admin can then see who has visited their network and gain insight into the OS and protocols they were using.

NIPDS (Network Intrusion and Prevention Detection System)

In NIPDS mode, SNORT will only log packets that are considered malicious. It does this using the preset characteristics of malicious packets, which are defined in its rules. The action that SNORT takes is also defined in the rules the network admin sets out.

What Are the Uses of SNORT Rules?

The rules defined in SNORT enable the software to carry out a range of actions, which include:

Perform Packet Sniffing

SNORT can be used to carry out packet sniffing, which collects all data that transmits in and out of a network. Collecting the individual packets that go to and from devices on the network enables detailed inspection of how traffic is being transmitted.

Debug Network Traffic

Once it has logged traffic, SNORT can be used to debug malicious packets and any configuration issues.

Generate Alerts

SNORT generates alerts to users as defined in the rule actions created in its configuration file. To receive alerts, SNORT rules need to contain conditions that define when a packet should be

considered unusual or malicious, the risks of vulnerabilities being exploited, and may violate the organization's security policy or pose a threat to the network.

Create New Rules

SNORT enables users to easily create new rules within the software. This allows network admins to change how they want SNORT conversion to work for them and the processes it should carry out. For example, they can create new rules that tell SNORT to prevent backdoor attacks, search for specific content in packets, show network data, specify which network to monitor, and print alerts in the console.

Differentiate Between Normal Internet Activities and Malicious Activities

Using SNORT rules enables network admins to easily differentiate between regular, expected internet activity and anything that is out of the norm. SNORT analyzes network activity in real time to sniff out malicious activity, then generates alerts to users.